

Areg Hovumyan

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Education

California State Polytechnic University, Pomona

Pomona, CA

Bachelor of Science in Computer Engineering — GPA: 3.6/4.0

Expected Graduation Jun 2028

Experience

Electrical/Embedded Systems Member: L'SPACE (Internship/Academy)

Remote

NASA's student acceleration program - Robotic Lunar Rover Mission

May 2026 – Present

- Presented the Communication and Data Handling to NASA engineers, got feedback, and improved the tools being used by choosing the right software based on their advice.
- Managed to stay within the schedule, cost, and mass limitations (under 250M, 170Kg, no sunlight for 14 days and more) for a possible mission on the moon.
- Designed the whole rover's Communication and Data Handling(C&DH) architecture by communicating with the mechanical and electrical team.
- Established (on concept) the communication between the Earth and the Moon, autonomous mission completion in case of redundancy using NeoPan, learned F' to fit the mission under the mission needs.

Software Engineer: Autonomous Vehicle Lab (Research)

Pomona, CA

University Research Lab

April 2026 – Present

- Create an interceptor drone, running object detection on a Jetson, establishing autonomous flight through a ready-made flight controller, and fitting under the budget of \$2000.
- Build a closed-loop stepper controller (CL57T) on a microcontroller with CAN telemetry over a TJA1051 transceiver to control a steering wheel through a computer, controller, or a joystick.
- Engineer CI/CD pipeline in GitHub Actions for the team to catch regressions before merge, and create unit tests to improve system reliability.
- Resolve priority-inversion bug by introducing mutex priority inheritance, restoring deterministic timing.
- Utilize AI agents to interface an MPU IMU over I2C on a Nucleo to stream 9-DOF telemetry.

Projects

Software Engineering Team Lead: Autonomous Hexacopter (University Club)

Pomona, CA

Lockheed Martin sponsored project

June 2025 – Present

- Apply pair programming, test-first development, unit testing, create containers for a ROS2 environment with Docker for a team of 8 student-engineers to collaborate, and create CI/CD pipelines to ship features with fewer defects.
- Currently integrating the hexacopter into Anduril's Lattice via a ROS2 bridge node that fuses MAVLink flight state with onboard perception, georeferencing pixel detections to WGS-84 ground coordinates and publishing them as C2 entities on the flight control loop.
- Collaborate with the Electrical team to not overwhelm the processor with object detection and image capturing processes, and work with the Payload team to drop payloads mid-flight using servos.
- Present mapping and Computer vision improvement, image processing achievement, and correct GPU/CPU utilization to Lockheed Martin engineers during the Preliminary Design Review (PDR).
- Create ROS2 nodes that handle the whole mission, including object detection (0.99 mAP@0.5), area mapping, 4k image processing (under 0.7s each), and payload release over a target.

Embedded Systems Member: Autonomous Ground Vehicle (University Club)

Pomona, CA

Northrop Grumman Collaboration Project

February 2026 – June 2026

- Present telemetry connection between a drone, rover, and a ground station workflow to Northrop Grumman engineers during a Preliminary Design Review (PDR).
- Implement Feature-driven development and the lean approach to release new code every week and meet all the requirements before any demonstrations or meetings.
- Run an authorized hardware review of a DS2431-class authenticator with a Bus Pirate v4 and logic analyzer, confirming a family-code mismatch and identifying clone silicon.
- Collaborate with the firmware team to develop bare-metal firmware for a pre-made microcontroller in C, directly manipulating registers (GPIO, timers, interrupts, etc.).

Skills

Languages and Tools: C, C++, Python, Rust, Java, VHDL, ARM Assembly, Bash, SQL, bare-metal, UART, SPI, ROS2 (Humble/Iron), Gazebo, YOLO, PyTorch, computer vision, machine learning, Linux, Git, GitHub Actions, Docker, MongoDB, Javascript, Typescript, Node.

Coursework: Microcontrollers & Embedded Systems, Circuit Analysis, Signals & Systems, Data Structures & Algorithms.

Languages: English, Russian, Armenian